

Lesson 11: Power Series Solutions - Leibniz Method

Professor Dave Explains Differential Equations Master Series · Lesson 11

Video: Power Series Solutions Part 1: Leibniz Method Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that when differential equations have variable coefficients (like Airy's equation $y'' - xy = 0$ in quantum mechanics), polynomial series expansions unlock exact solutions. By expressing the unknown solution as an infinite power series $y(x) = \sum c_n x^n$, differentiating term-by-term, and shifting summation indices, the differential equation transforms into an algebraic recurrence relation that generates every coefficient in terms of the initial conditions $c_0 = y(0)$ and $c_1 = y'(0)$.

2. Core Theoretical Framework & Rigorous Formulations

Power Series Algorithm around Ordinary Point $x_0 = 0$

- 1. **Series Expansion:** $y(x) = \sum_{n=0}^{\infty} c_n x^n$, $y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}$, $y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$.
- 2. **Index Shifting:** Shift indices to align powers of x^k . In y'' , let $k = n - 2 \implies n = k + 2$:
$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} = \sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$$
- 3. **Combine and Factor:** Group all sums under a single $\sum_{k=0}^{\infty} [\dots] x^k = 0$.
- 4. **Recurrence Relation:** Set the bracketed coefficient to zero to express c_{k+2} in terms of earlier coefficients.
- 5. **Decompose into Independent Bases:** $y(x) = c_0 y_1(x) + c_1 y_2(x)$.

3. Fully Solved Whiteboard Problems

Concordia Final Exam Problem: Power Series for Airy-Type ODE

Problem Statement: Find the power series solution of $y'' - xy = 0$ about the ordinary point $x_0 = 0$ up to terms in x^5 .

- Step 1: Substitute Series Expansions**
 $\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n = 0 \implies \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} = 0$.
- Step 2: Align Exponents to x^k**
For 1st sum: $k = n - 2 \implies \sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$. For 2nd sum: $k = n + 1 \implies \sum_{k=1}^{\infty} c_{k-1} x^k$.
- Step 3: Peel Off $k = 0$ Term and Combine for $k \geq 1$**
At $k = 0$: $(2)(1) c_2 = 0 \implies c_2 = 0$. For $k \geq 1$: $(k+2)(k+1) c_{k+2} - c_{k-1} = 0 \implies c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}$.
- Step 4: Compute Higher Coefficients**
For $k = 1$: $c_3 = \frac{c_0}{(3)(2)} = \frac{c_0}{6}$. For $k = 2$: $c_4 = \frac{c_1}{(4)(3)} = \frac{c_1}{12}$. For $k = 3$: $c_5 = \frac{c_2}{(5)(4)} = 0$ (since $c_2 = 0$).
- Step 5: Assemble General Solution**
 $y(x) = c_0 \left(1 + \frac{x^3}{6} + \dots \right) + c_1 \left(x + \frac{x^4}{12} + \dots \right)$.

 **Common Exam Pitfall & Trapdoor Avoidance**

Always peel off mismatched low-order terms (like $k = 0$) before combining summations into a single bracket. Forgetting to set isolated terms to zero causes coefficient chaos.

4. Key Takeaways & Exam Strategy Formula Card

 **Core Takeaway & Exam Rule**

Ordinary points always yield two linearly independent analytic series $y_1(x)$ (even/odd branches) weighted by c_0 and c_1 .

- Video reference: [Power Series Solutions Part 1: Leibniz Method](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.